

2023 Ch H2 Q6

Section: Chemical Changes and Structure

Topic: Equilibria, Rates and Energy /
Industrial Processes

Question Summary

This question examines ammonium nitrate as a fertiliser, the Haber process, enthalpy diagrams, excess reagent calculations, rate/energy profiles, equilibrium concepts, and ammonia in fuel cells.

Worked Solution

According to the SQA Marking Instructions for 2023 H2 Q6:

(a)(i)(A) The enthalpy diagram should show products at lower energy than reactants, since ΔH is negative (exothermic). The curve has a peak representing activation energy. (B) The species at X is the activated complex.

(a)(ii) Calculate reactant amounts:

- HNO_3 : $1316 \text{ L} \times 9.5 \text{ mol L}^{-1} = 12,502 \text{ mol}$.

- NH_3 : $220,000 \text{ g} \div 17 = 12,941 \text{ mol}$.

Reaction ratio 1:1, so NH_3 is in slight excess (12,941 vs 12,502).

(a)(iii) The direct neutralisation route ($\text{NH}_3 + \text{HNO}_3$) is preferred as it avoids by-products such as CaCO_3 and is more efficient/cheaper.

(b)(i)(A) The area under the Maxwell-Boltzmann distribution curve represents the total number of molecules.

(B) At higher temperature, the curve shifts: peak lower and to the right, broader tail.

(b)(ii) On Graph 2, the activation energy line moves to the left when a catalyst is added (lower E_a).

(c)(i) Dynamic equilibrium: the forward and reverse reactions occur at equal rates in a closed system, so concentrations remain constant.

(c)(ii)(A) Continuously removing ammonia shifts equilibrium to the right (Le Chatelier's principle), increasing yield.

(B) From the flow diagram: recycling unreacted gases reduces waste and cost, maximising profit and minimising environmental impact.

(d)(i) Overall redox reaction in fuel cell:
 $2\text{NH}_3 + 3\text{O}_2 \rightarrow \text{N}_2 + 6\text{H}_2\text{O}$.

(ii) The reducing agent is ammonia (loses electrons).

Final Answer

(a)(i) Exothermic diagram; X = activated complex.

(a)(ii) NH_3 in excess.

(a)(iii) Preferred as it avoids by-products, more efficient.

**(b)(i) (A) Area = total molecules.
(B) Curve lower, broader, shifted right.**

(b)(ii) Catalyst lowers E_a , line moves left.

(c)(i) Dynamic equilibrium = forward & reverse at equal rate, closed system.

(c)(ii) (A) Removing NH_3 shifts equilibrium right. (B) Recycling gases reduces waste & cost.

(d)(i) $2\text{NH}_3 + 3\text{O}_2 \rightarrow \text{N}_2 + 6\text{H}_2\text{O}$. (ii) Reducing agent = NH_3 .

Revision Tips

- Activated complex = unstable arrangement at E_a peak.
- Compare moles to check limiting reagent.
- Maxwell-Boltzmann: higher T = more particles with $E > E_a$.
- Catalysts lower E_a but don't change ΔH .
- Equilibrium shifts to oppose change (Le Chatelier).

- Recycling & heat exchangers improve efficiency in industry.





